

Abdelrahman Elsayed Ali Hassan Shaban

Data Scientist | Machine Learning, Deep Learning & Data Engineering

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github.com/Abdelrahman-0-20 | [portfolio: main-website-profile-abdelrahman-0-20.vercel.app](#) | share.streamlit.io/user/abdelrahman-0-20

PROFESSIONAL SUMMARY

Data Scientist with a Computer Science background and hands-on experience across the full machine learning and analytics lifecycle — from classical ML and deep learning (CNNs, transfer learning) to interactive BI dashboards and production-style deployment (FastAPI, Docker, Streamlit). Skilled in Python, SQL, and Power BI, with experience turning raw, messy data into predictive models and stakeholder-ready insights. Trained through intensive, project-based programs at Epsilon AI Academy, ALX Africa, and ExploreAI. Combines strong quantitative reasoning with clear technical communication to deliver measurable impact.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL (CTEs, Window Functions, Complex Joins; Microsoft SQL Server, PostgreSQL), Excel (Advanced Formulas, PivotTables, Power Query)
- **Machine Learning & Deep Learning:** scikit-learn (Linear/Logistic Regression, Random Forest, KNN, K-Means, PCA), Deep Learning (CNNs, Transfer Learning, ResNet architectures), Feature Engineering (StandardScaler, One-Hot Encoding, TF-IDF), Model Evaluation (Cross-Validation, ROC-AUC, RMSE, F1-score)
- **MLOps & Deployment:** FastAPI, Docker, Streamlit (ML App Deployment), Git/GitHub, AWS Cloud Fundamentals (EC2, S3)
- **Visualization & BI:** Power BI (Data Modeling, DAX Measures, Interactive Dashboards), Google Sheets
- **Core Competencies:** Exploratory Data Analysis (EDA), NLP, Statistical Hypothesis Testing, KPI Tracking & Reporting, Data Storytelling, Object-Oriented Programming (OOP), ETL Pipelines

SELECTED PROJECTS

Image Classification — ResNet Transfer Learning Pipeline

- Built a CNN image classification pipeline using ResNet-based transfer learning.
- Identified and fixed a critical data transform-leakage bug, and added mixed precision training, stratified splits, and cosine annealing to improve reliability and training efficiency.

Fraud Detection Pipeline — FastAPI & Docker

- Built and containerized a fraud detection service using FastAPI and Docker for deployment.
- Trained a Random Forest Classifier achieving an ROC-AUC of ~0.94 on transaction data.

Customer Segmentation — K-Means & PCA

- Applied K-Means clustering and PCA on RFM (Recency, Frequency, Monetary) features to segment customers.
- Delivered 5 business-labeled customer segments to support targeted marketing strategy.

Sentiment Analysis — NLP & Neural Networks

- Built a sentiment classification model on the IMDb movie review dataset using TF-IDF features and a neural network classifier to predict review polarity.

Spam Filter — SMS Text Classification

- Engineered an end-to-end NLP pipeline with scikit-learn and NLTK, applying TF-IDF vectorization with (1,2) n-grams.
- Tuned a Multinomial Naive Bayes classifier to achieve an F1-score above 0.95.

Breast Cancer Classification Model

- Built a Random Forest classifier on the Wisconsin Breast Cancer dataset with StandardScaler normalization and GridSearchCV tuning, achieving 92% accuracy and near-perfect recall.

House Price Prediction — Regression Web App

- Performed EDA on 60,000 property records, identifying sqft_living, zipcode, and grade as top price drivers.
- Built and deployed a Streamlit regression app achieving a test R^2 of 0.88 versus 0.72 for a baseline Linear Regression model.

TRAINING & PRACTICAL EXPERIENCE

Data Science & Analytics Portfolio Projects | 2025 – Present

- Designed and deployed 3 Power BI dashboards spanning house sales and customer analytics scenarios.
- Built DAX measures for a Customer Personality Analysis dashboard, segmenting customers by purchasing behavior and identifying a high-value cohort representing 20% of total simulated revenue.

Data Analysis Professional (CDAP) — Epsilon AI Academy Egypt | Mar 2025 | 110 Hours

- Led the full analytics lifecycle in a capstone project: raw data acquisition, advanced Excel modeling (XLOOKUP, INDEX-MATCH, What-If Analysis), and an interactive Power BI executive dashboard.

Professional Diploma in Data Science & Machine Learning — Mansoura University | 2024 – 2025 | 7 Months

- Built end-to-end Python pipelines to clean messy real-world datasets, including custom functions to detect and impute missing values and standardize outlier treatment.
- Developed and validated predictive models with scikit-learn: Linear Regression for continuous forecasting (R^2 , RMSE) and Logistic Regression/KNN for classification (cross-validation for optimal K).

Data Analytics Program — ExploreAI Academy & ALX Africa | Sep 2024 | 6 Months

- Mastered core data-driven decision-making skills: optimized SQL queries, advanced spreadsheet preparation (PivotTables, VLOOKUP), and Power BI storytelling.

Python Developer — Applied Projects & Coursework | 2024 – 2025

- Applied OOP principles to build a Support Ticket Management System automating categorization, priority assignment, and status tracking, with a Streamlit dashboard visualizing ticket volume and resolution times.

EDUCATION

B.S. in Computer Science — Higher Institute of Information Technology Management (HIMIT), Kafr El-Sheikh | 2022 – 2025

- Graduated with “Good+” overall standing and an “Excellent” grade on capstone graduation project — a full-cycle sentiment analysis system for movie reviews using ML and NLP.
- Relevant coursework: Database Management Systems (SQL, relational normalization 1NF–3NF, ACID properties), Object-Oriented Programming (Encapsulation, Inheritance, Polymorphism).

Foundation Year in Programming & Embedded Systems — Higher Future Institute for Engineering and Technology | 2021 – 2022

- Built a strong algorithmic foundation through C# fundamentals, data structures, and problem-solving techniques.
- Collaborated on an Arduino-based autonomous vehicle project spanning hardware-software integration.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Build AI Agents with n8n — Edureka
- Artificial Intelligence Technology Diploma — Mansoura University (Oct 2023, 96 Hours)
- ALX Professional Foundations & Founder Academy (May 2024) — Leadership & problem-solving frameworks
- Microsoft Excel Specialization — Coursera (Apr 2024)
- Google Data Analytics: Foundations — Google/Coursera (Apr 2024)
- SQL Manipulation (Intro & Intermediate) — DataCamp (Apr 2024)
- AWS Cloud Practitioner Fundamentals — DataCamp

LANGUAGES

- Arabic — Native
- English — Professional Working Proficiency (B2)